

Numeric methods for PDE

1. Diffusion IVP

$$u_t = au_{xx}$$

with,

$$\begin{aligned} u(0, t) &= g_0(t) \quad , \quad 0 \leq t \leq t_f \\ u(1, t) &= g_1(t) \quad , \quad 0 \leq t \leq t_f \\ u(x, 0) &= h(x) \quad , \quad 0 \leq x \leq 1 \end{aligned}$$

by applying forward difference to u_t and central difference to u_{xx} with $x_i = i\Delta x$ and $t_k = k\Delta t$

$$\frac{U(x_i, t_{k+1}) - U(x_i, t_k)}{\Delta t} = a \frac{U(x_{i+1}, t_k) - 2U(x_i, t_k) + U(x_{i-1}, t_k)}{\Delta x^2}$$

where $U(x_i, t_{k+1})$ is the approximation of $u(x_i, t_{k+1})$
So,

$$U(x_i, t_{k+1}) = U(x_i, t_k) + a \frac{\Delta t}{\Delta x^2} (U(x_{i+1}, t_k) - 2U(x_i, t_k) + U(x_{i-1}, t_k))$$

approximation at x_i, t_{k+1} depends on x_{i+1}, x_{i-1}, x_i at t_k

$$u(x_i, t_{k+1}) = u(x_i, t_k) + \Delta t u_t + O(\Delta t^2)$$

Hence

$$\frac{u(x_i, t_{k+1}) - u(x_i, t_k)}{\Delta t} = u_t + O(\Delta t)$$

Similarly,

$$\frac{u(x_{i+1}, t_k) - 2u(x_i, t_k) + u(x_{i-1}, t_k)}{\Delta x^2} = u_{xx} + O(\Delta x^2)$$

if we multiply the second equation above by a and subtract with first equation we can see that the truncation error will be of order $O(\Delta t) + O(\Delta x^2)$.

Global error is defined as $e(x_i, t_k) = U(x_i, t_k) - u(x_i, t_k)$

Subtracting the equations provided above

$$\frac{e(x_i, t_{k+1}) - e(x_i, t_k)}{\Delta t} = a \frac{e(x_{i+1}, t_k) - 2e(x_i, t_k) + e(x_{i-1}, t_k)}{\Delta x^2} - T$$

$$e(x_i, t_{k+1}) = \left(1 - 2a \frac{\Delta t}{\Delta x^2}\right) e(x_i, t_k) + a \frac{\Delta t}{\Delta x^2} e(x_{i+1}, t_k) + a \frac{\Delta t}{\Delta x^2} e(x_{i-1}, t_k) - T \Delta t$$

where T is the truncation error

Let $E_k = \max_{0 \leq i \leq N} |e(x_i, t_k)|$ and assume that $\Delta t \leq \frac{1}{2} \frac{\Delta x^2}{a}$ and $T_{\max}$ the maximum value of turncation error

$$|e(x_i, t_{k+1})| \leq \left| 1 - 2a \frac{\Delta t}{\Delta x^2} \right| E_k + \left| a \frac{\Delta t}{\Delta x^2} \right| E_k + \left| a \frac{\Delta t}{\Delta x^2} \right| E_k + \Delta t T_{\max}$$

absotot absolute value can be removed due to above assumption

$$|e(x_i, t_{k+1})| \leq E_k + \Delta t T_{\max}$$

with $|e(x_0, t_{k+1})| = |e(x_N, t_{k+1})| = 0$

$$E_{k+1} \leq E_k + \Delta t T_{\max}$$

$$E_1 \leq \Delta t T_{\max}$$

$$E_2 \leq 2\Delta t T_{\max}$$

$$\vdots$$

$$E_n \leq n\Delta t T_{\max}$$

let $n\Delta t \leq t$

$$E_n \leq t T_{\max}$$

So as long as $\Delta t \leq \frac{1}{2} \frac{\Delta x^2}{a}$ the error is bounded

An implicit method would be

$$\frac{U(x_i, t_{k+1}) - U(x_i, t_k)}{\Delta t} = a \frac{U(x_{i+1}, t_{k+1}) - 2U(x_i, t_{k+1}) + U(x_{i-1}, t_{k+1}))}{\Delta x^2}$$

Let $r = a \frac{\Delta t}{\Delta x^2}$

$$U(x_i, t_{k+1}) = U(x_i, t_k) + rU(x_{i+1}, t_{k+1}) - 2rU(x_i, t_{k+1}) + U(x_{i-1}, t_{k+1})$$

$$(1 + 2r)U(x_i, t_{k+1}) - rU(x_{i+1}, t_{k+1}) - rU(x_{i-1}, t_{k+1}) = U(x_i, t_k)$$

point t_k depends on three other points forward in time .

Let

$$U^0 = \begin{bmatrix} h(x_1) \\ h(x_2) \\ \vdots \\ h(x_{N-1}) \end{bmatrix} \quad \text{and} \quad U^{k+1} = \begin{bmatrix} U(x_1, t_{k+1}) \\ U(x_2, t_{k+1}) \\ \vdots \end{bmatrix}$$

$$\text{and } b^{k+1} = \begin{bmatrix} rg_0(t_{k+1}) \\ 0 \\ \vdots \\ rg_1(t_{k+1}) \end{bmatrix} \text{ and } A = \begin{bmatrix} 1+2r & -r & 0 & \dots & \dots & 0 \\ -r & 1+2r & -r & & & \\ 0 & -r & 1+2r & -r & & \\ \vdots & & \ddots & \ddots & \ddots & \\ 0 & \dots & 0 & -r & 1+2r & \end{bmatrix}$$

the the set of equations from the scheme can be put as

$$AU^{k+1} = U^k + b^{k+1}$$

The truncation error

$$T(x_i, t_{k+1}) = \frac{u(x_i, t_{k+1}) - u(x_i, t_k)}{\Delta t} - a \frac{u(x_{i+1}, t_{k+1}) - 2u(x_i, t_{k+1}) + u(x_{i-1}, t_{k+1}))}{\Delta x^2}$$

from tylor expansion we can deduce that the truncation error is of the order $O(\Delta t) + O(\Delta x^2)$

The error equation would be for the form

$$Ae^{k+1} = e^k + \Delta t T^{k+1}$$

where T^{k+1} is the vector of truncation error at $k+1$ th time point . Since A is tridiagonal if $r > 0$ then

$$\|A^{-1}\|_{\infty} \leq 1$$

$$\|e^{k+1}\|_{\infty} \leq \|e^k\|_{\infty} + \Delta t \|T\|_{\infty}$$

with $e^0 = 0$

$$\|e^n\|_{\infty} \leq t T_{\max}$$

for this method there was no assumption like $\Delta t \leq \frac{1}{2} \frac{\Delta x^2}{a}$.

2. Transport equation

In its general form the one-dimensional transport equation ,

$$u_t = -v(x, t)u_x + a(x, t)u + f(x, t)$$

$$u(0, t) = g(t) \quad 0 \leq t \leq t_f$$

$$u(x, 0) = h(x) \quad 0 \leq x \leq 1$$

Using forward differin in space won't make sense due to the domain of dependence (value at x depends on value at $x - vt$ only for a constant v) . Likewise for central difference.Hence the follwing scheme,

$$\frac{U(x_i, t_{k+1}) - U(x_i, t_k)}{\Delta t} = -v(x_i, t_k) \frac{U(x_i, t_k) - U(x_{i-1}, t_k)}{\Delta x} + a(x_i, t_k)U(x_i, t_k) + f(x_i, t_k)$$

with ,

$$U(x_0, t_k) = g(t_k)$$

$$U(x_i, t_0) = h(x_i)$$

this scheme only depends on space points to the "left"

Since

$$\frac{u(x_i, t_{k+1}) - u(x_i, t_k)}{\Delta t} = u_t + \frac{\Delta t}{2} u_{tt} + O(\Delta t^2)$$

$$v(x_i, t_k) \frac{u(x_i, t_k) - u(x_{i-1}, t_k)}{\Delta x} = v(x_i, t_k) u_x - a \frac{\Delta x}{2} u_{xx} + O(\Delta x^2)$$

So truncation error will be

$$T(x_i, t_k) = \frac{1}{2} \Delta t u_{tt} - \frac{1}{2} \Delta x v(x_i, t_k) u_{xx} = O(\Delta t) + O(\Delta x)$$

The equation for error $e = U(x_i, t_k) - u(x_i, t_k)$ is obtained by subtracting the equation

$$\frac{e(x_i, t_{k+1}) - e(x_i, t_k)}{\Delta t} = -v(x_i, t_k) \frac{e(x_i, t_k) - e(x_{i-1}, t_k)}{\Delta x} + a(x_i, t_k) e(x_i, t_k) - T(x_i, t_k)$$

with,

$$e(x_0, t_k) = 0$$

$$e(x_i, t_0) = 0$$

$$e(x_i, t_{k+1}) = \left[1 - \frac{\Delta t}{\Delta x} v(x_i, t_k) + \Delta t a(x_i, t_k) \right] e(x_i, t_k) + \frac{\Delta t}{\Delta x} v(x_i, t_k) e(x_{i-1}, t_k) - \Delta t T(x_i, t_k)$$

let

$$E_k = \max_{0 \leq i \leq N} |e(x_i, t_k)|$$

$$T_{\max} = \max |T(x_i, t_k)|$$

$$A_{\max} = \max |a(x, t)|$$

$$V_{\max} = \max |v(x, t)|$$

$$|e(x_i, t_{k+1})| \leq \left| 1 - \frac{\Delta t}{\Delta x} v(x_i, t_k) \right| E_k + \frac{\Delta t}{\Delta x} |v(x_i, t_k)| E_k + \Delta t A_{\max} E_k + \Delta t T_{\max}$$

to remove the absolute value $1 \geq \frac{\Delta t}{\Delta x} v_{\max}$,
with this condition the equation will become

$$\begin{aligned} |e(x_i, t_{k+1})| &\leq (1 + \Delta t A_{\max}) E_k + \Delta t T_{\max} \\ &\leq (1 + \Delta t A_{\max}) [(1 + \Delta t A_{\max}) E_{k-1} + \Delta t T_{\max}] + \Delta t T_{\max} \\ &= (1 + \Delta t A_{\max})^2 E_{k-1} + [(1 + \Delta t A_{\max}) + 1] \Delta t T_{\max} \\ &\vdots \\ &\leq (1 + \Delta t A_{\max})^{k+1} E_0 + [(1 + \Delta t A_{\max})^k + \dots + (1 + \Delta t A_{\max}) + 1] \Delta t T_{\max} \end{aligned}$$

Since

$$1 \leq (1 + hA) \leq (1 + hA)^2 \leq \dots \leq (1 + hA)^k \leq (1 + hA)^{k+1}$$

if $h > 0$, $A \geq 0$, since Δt and $A_{\max}$ satisfy this condition

$$|e(x_i, t_{k+1})| \leq (1 + \Delta t A_{\max})^{k+1} (E_0 + (k+1) \Delta t T_{\max})$$

$$E_0 = 0 \quad \text{and} \quad (k+1) \Delta t = t_{k+1}$$

$$\text{and } (1 + \Delta t A_{\max})^{k+1} \leq \exp(A_{\max} t_{k+1})$$

Hence

$$E_k \leq \exp(A_{\max} t_k) t_k T_{\max}$$

$$\text{when } \Delta t \leq \frac{\Delta x}{v_{\max}}$$

This condition is needed because the domain of dependence of the approximation must contain the point $x - vt$ (assuming v is constant here)

3. Wave Equation

$$u_{tt} = c^2 u_{xx} + f(x, t)$$

$$u(0, t) = g_0(t) \quad , \quad u(x, 0) = h(x)$$

$$u(1, t) = g_1(t) \quad , \quad u_t(x, 0) = q(x)$$

applying central difference for space and time

$$\frac{U(x_i, t_{k+1}) - 2U(x_i, t_k) + U(x_i, t_{k-1}))}{\Delta t^2} = c^2 \frac{U(x_{i+1}, t_k) - 2U(x_i, t_k) + U(x_{i-1}, t_k)}{\Delta x^2} + f(x_i, t_k)$$

$$U(x_0, t_k) = g_0(t_k)$$

$$U(x_N, t_k) = g_1(t_k)$$

$$U(x_i, t_0) = h(x_i)$$

To handle velocity condition ,,

$$U(x_i, \Delta t) = U(x_i, 0) + \Delta t u_t(x_i, 0) + \frac{\Delta t^2}{2} u_{tt}(x_i, 0) + O(t^3)$$

Since

$$\begin{aligned} U(x_i, 0) &= h(x_i) \implies U_{xx}(x_i, 0) = h''(x_i) \\ U_t(x_i, 0) &= q(x_i) \\ u_{tt} &= c^2 u_{xx} + f \\ u_{tt}(x_i, 0) &= c^2 h''(x_i) + f(x_i, 0) \end{aligned}$$

Therefore

$$U(x_i, t_1) = h(x_i) + \Delta t q(x_i) + \frac{\Delta t^2}{2} [c^2 h''(x_i) + f(x_i, 0)]$$

Can be used in the initial condition. (to determine a point at t_1 we need data from two previous time steps)

$$\begin{aligned} \frac{u(x_i, t_{k+1}) - 2u(x_i, t_k) + u(x_i, t_{k-1}))}{\Delta t^2} &= u_{tt} + \frac{\Delta t^2}{12} u_{tttt} + O(\Delta t^4) \\ c^2 \frac{u(x_{i+1}, t_k) - 2u(x_i, t_k) + u(x_{i-1}, t_k))}{\Delta x^2} &= c^2 u_{xx} + c^2 \frac{\Delta x^2}{12} u_{xxxx} + O(\Delta x^4) \end{aligned}$$

Multiplying & substituting this in truncation error formula we get

$$T_i^k = \left(u_{tt} + \frac{\Delta t^2}{12} u_{tttt} \right) - c^2 \left(u_{xx} + \frac{\Delta x^2}{12} u_{xxxx} \right) - f + O(\Delta t^2) + O(\Delta x^2)$$

since $u_{tt} - c^2 u_{xx} - f = 0$, the order of truncation error is $O(\Delta t^2) + O(\Delta x^2)$

The equation satisfied by the error

$$\frac{e(x_i, t_{k+1}) - 2e(x_i, t_k) + e(x_i, t_{k-1}))}{\Delta t^2} = c^2 \frac{e(x_{i+1}, t_k) - 2e(x_i, t_k) + e(x_{i-1}, t_k))}{\Delta x^2} - T_i^k$$

with

$$\begin{aligned} e(x_0, t_k) &= 0 \\ e(x_N, t_k) &= 0 \\ e(x_i, t_0) &= 0 \\ e(x_i, t_1) &= -\frac{1}{6} \Delta t^3 u_{ttt}(x_i, \xi_i) \end{aligned}$$

Fourier Stability Method is needed to analyse this;
consider DFT

$$\begin{aligned} e(x_i, t_k) &= \sum_{m=1}^{N-1} a_m(t_k) \sin(m\pi x_i) \\ T_i^k &= \sum_{m=1}^{N-1} b_m(k) \sin(m\pi x_i) \end{aligned}$$

So,

$$\frac{e(x_i, t_{k+1}) - 2e(x_i, t_k) + e(x_i, t_{k-1}))}{\Delta t^2} = \sum_m \frac{1}{\Delta t^2} [a_m(t_{k+1}) - 2a_m(t_k) + a_m(t_{k-1}))] \sin(m\pi x_i)$$

Using

$$\sin(m\pi x_{i+1}) + \sin(m\pi x_{i-1}) = 2 \sin(m\pi x_i) \cos(m\pi \Delta x)$$

$$\frac{e(x_{i+1}, t_k) - 2e(x_i, t_k) + e(x_{i-1}, t_k))}{\Delta x^2} = \sum_m \left[\frac{2 \cos(m\pi \Delta x) - 2}{\Delta x^2} a_m(t_k) \sin(m\pi x_i) \right]$$

Substituting all of these back into error equation & comparing coefficients we get

$$\frac{a_m(t_{k+1}) - 2a_m(t_k) + a_m(t_{k-1}))}{\Delta t^2} = c^2 \frac{2 \cos(m\pi \Delta x) - 2}{\Delta x^2} a_m(t_k) - b_m(k)$$

or

$$a_m(t_{k+1}) + \left[-2 + 4c^2 \frac{\Delta t^2}{\Delta x^2} \sin^2 \left(\frac{1}{2} m\pi \Delta x \right) \right] a_m(t_k) + a_m(t_{k-1}) = -\Delta t^2 b_m(k)$$

Now Consider using the following theorem
if

$$c_{k+1} + \rho_1 c_k + \rho_2 c_{k-1} + \dots + \rho_m c_{k+1-m} = h T_k \quad (k \geq m-1)$$

and roots of $\lambda^m + \rho_1 \lambda^{m-1} + \dots + \rho_m$ are all $|\lambda_i| \leq 1$

then

$$|c_k| \leq M_\rho [\max(|c_0|, \dots, |c_{m-1}|) + t_k T]$$

(where $t_k = kh$ and $T = \max |T_j|$)

Since

$$\lambda^2 + \left[-2 + 4c^2 \frac{\Delta t^2}{\Delta x^2} \sin^2 \left(\frac{1}{2} m\pi \Delta x \right) \right] \lambda + 1$$

will have roots whose absolute value equal to one

When $\Delta t \leq \frac{\Delta x}{c}$ the theorem can be applied

Hence

$$|a_m(t_k)| \leq M_\rho \left[\max(|a_m(t_0)|, |a_m(t_1)|) + t_k \Delta t \max_{j \leq k} |b_m(j)| \right]$$

Using inverse fourier transform we have

$$b_l(j) = \frac{2}{N} \sum_{i=1}^{N-1} T_i^j \sin(l\pi x_i) \implies |b_l(j)| \leq \frac{2}{N} (N-1) T_{\max} \leq 2 T_{\max}$$

$$a_l(t_0) \leq \frac{2}{N}(N-1)E_0 = 0$$

$$a_l(t_1) \leq \frac{2}{N}(N-1)E_1 \leq 2E_1$$

where E_k is same as before

Hence

$$|a_m(t_k)| \leq M_\rho(2E_1 + t_k \Delta t 2T_{\max})$$

$$|e(x_i, t_k)| \leq (N-1) \max |a_m(t_k)|$$

$$\leq \frac{\Delta t}{\Delta x} M_\rho \left[\frac{2E_1}{\Delta t} + 2t_k T_{\max} \right]$$

So Finally

$$E_k \leq \frac{M_\rho}{c} \left[\frac{2E_1}{\Delta t} + 2t_k T_{\max} \right]$$

For the scheme given when $\Delta t \leq \frac{\Delta x}{c}$
and since Maximum error at t_1 is $O(\Delta t^3)$
the overall error will be $O(\Delta x^2) + O(\Delta t^2)$ (from above expression).